

SS 465 Data from carpenter technology

REFERENCE LINK

<https://www.carpentertechnology.com/hubfs/7407324/Material%20Saftey%20Data%20Sheets/Custom%20465.pdf>

DENSITY

Properties	
Physical Properties	
Density	
Annealed/CT	0.2822 lb/in ³
Condition H 900	0.2825 lb/in ³
Condition H 950	0.2829 lb/in ³
Condition H 1000	0.2832 lb/in ³
Condition H 1050	0.2832 lb/in ³
Condition H 1100	0.2840 lb/in ³
Density	
Condition	g/cm ³
Annealed/CT	7.81
H900	7.82
H950	7.83
H1000	7.84
H1050	7.84
H1100	7.86

YOUNG'S MODULUS

CarTech® Custom 465® Stainless	
Modulus of Elasticity (E)	
Condition H 1000	28.8 x 10 ³ ksi
Condition H 1100	28.4 x 10 ³ ksi
Modulus of Elasticity (E)	
Condition	X10 ³ ksi
H1000	28.8
H1100	28.4

TEMP DEPENDENT THERMAL CONDUCTIVITY

Thermal conductivity					
Temperature Range		W/m•K (BTU•in/ft ² •h•F)			
°F	°C	Annealed/CT	H900	H1000	H1050
73	23	14.06 (97.6)	14.85 (103.0)	15.83 (109.8)	15.80 (109.6)
212	100	15.65 (108.6)	16.89 (117.2)	18.01 (125.0)	18.09 (125.5)
392	200	17.71 (122.9)	19.21 (133.3)	20.25 (140.5)	20.42 (141.7)
572	300	19.37 (134.4)	20.97 (145.5)	21.71 (150.6)	21.99 (152.6)
752	400	21.43 (148.7)	22.49 (156.0)	23.55 (163.4)	23.51 (163.1)
932	500	23.85 (165.5)	24.52 (170.1)	25.58 (177.5)	25.29 (175.5)
1112	600	25.76 (178.7)	26.32 (182.6)	27.61 (191.6)	26.91 (186.7)

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MEAN COEFFICIENT OF THERMAL EXPANSION

CarTech® Custom 465® Stainless

Mean coefficient of thermal expansion

Temperature Range		Coefficient of Expansion ($10^{-6}/^{\circ}\text{C}$)			
$^{\circ}\text{F}$	$^{\circ}\text{C}$	Annealed/CT	H900	H1000	H1100
77/212	25/100	10.30	10.40	10.60	11.30
77/392	25/200	10.80	11.10	11.10	12.00
77/572	25/300	10.90	11.40	11.50	12.40
77/752	25/400	11.10	11.70	11.70	12.70
77/932	25/500	10.90	12.00	12.00	12.90
77/1112	25/600	9.86	11.20	12.20	13.10